

Credit Risk Prediction

Give Me Some Credit | One-Page Project Summary

Project Background

Predict serious delinquency within two years using Kaggle borrower records, and translate predicted probabilities into simulated credit approval decisions.

Data and Cleaning

The raw dataset contained 150,000 records and 12 columns. Cleaning removed the row index, 1 invalid-age record(s), 269 abnormal-code records and 565 duplicates. 149,165 records remained. Income missingness was retained; imputation and debt capping were learned from the fitting partition.

Methods and Metrics

A stratified 75%/25% split was used. Logistic Regression and LightGBM were compared. Calibrated LightGBM validation AUC was 0.8592, KS was 0.5691, and Brier score was 0.0505. Five-fold training-set LightGBM CV AUC was 0.8637 (standard deviation 0.0022). Sigmoid calibration used training-only folds.

Strategy Conclusion

The highest-risk 10% had a target rate of 35.47%, versus 6.62% overall (lift 5.36). Assuming RMB 1,000 profit per good loan and RMB 7,000 loss per bad loan, the 10% cutoff gave the highest retrospective simulated profit among four tested cutoffs: RMB 24,365,000, with 90.09% approved and 3.43% observed delinquency among approved borrowers. This validation-set result requires independent confirmation.

Tools

Python, pandas, NumPy, matplotlib, scikit-learn, LightGBM, joblib, and Kaggle Notebook.

